

Lab2 Part C Q1

```
Code:
#include <iostream>
#include <string>
using namespace std;

bool checkInput(string time); int
processElapsedTime(string time);

int main()
{
    string time;

    cout << "Please enter your elapsed time (in HH:MM:SS format) = ";
    cin >> time;

    while (checkInput(time) == false)
    {
        cout << "Invalid time! Please enter again (in HH:MM:SS format) = ";
        cin >> time;
    }
    int seconds = processElapsedTime(time);

    cout << "Elapsed time in seconds = " << seconds << " seconds" << endl;

    return 0;
}
bool checkInput(string time)
{
    if (time.length() != 8)
    {
        return false;
    }
    if (time[2] != ':' || time[5] != ':')
    {
        return false;
    }
    for (int i = 0; i <
8; i++)
    {
        if (i != 2 && i != 5)
        {
            if (time[i] < '0' || time[i] > '9')
            {
                return false;
            }
        }
    }
    int hour = (time[0] - '0') * 10 + (time[1] - '0');
    int minute = (time[3] - '0') * 10 + (time[4] - '0');
    int second = (time[6] - '0') * 10 + (time[7] - '0');
    if (hour > 23 || minute > 59 || second >
59)
    {
        return false;
    }
}
```

```

        return true;
    }
    int processElapsedTime(string time)
    {
        int hour = (time[0] - '0') * 10 + (time[1] - '0');
        int minute = (time[3] - '0') * 10 + (time[4] - '0');
        int second = (time[6] - '0') * 10 + (time[7] - '0');

        int total = hour * 3600 + minute * 60 + second;

        return total; }

```

Output:

```

PS C:\Users\ianga\Downloads\Data Structure\C++ Codes> & "C:\Users\ianga\Downloads\Data Structure\C++ Codes\Lab2_PartC_Q1.exe"
Please enter your elapsed time (in HH:MM:SS format) = 13:18:20
Elapsed time in seconds = 47900 seconds

```

Lab 2 Part C Q2

```

#include <iostream>
#include <string>
using namespace std;
    string ConvertLetterString(string month);
    int FindDaysInMonth(string month, int
year);
    void weatherReport(int days, int &hot, int &rainy, int &cloudy);
    void displayReport(int hot, int rainy, int cloudy);

int main()
{
    string
month;    int
year;    int
days;
    int hot = 0, rainy = 0, cloudy = 0;

    cout << "Enter Your Month (e.g. August 2019): ";
    cin >> month >> year;
    month =
ConvertLetterString(month);    days =
FindDaysInMonth(month, year);

    while (days == 0)
    {
        cout << "Invalid month! Enter Your Month (e.g. August 2019): ";
        cin >> month >> year;
        month = ConvertLetterString(month);
        days = FindDaysInMonth(month, year);    }
    weatherReport(days, hot, rainy,
cloudy);    displayReport(hot, rainy,
cloudy);

```

```
        return 0;
    }
    string ConvertLetterString(string month)
    {
        string newMonth = "";

        for (int i = 0; i < month.length(); i++)
        {
            char c = month[i];

            if (i == 0)
            {
                if (c >= 'a' && c <= 'z')
                {
                    c = c - 32;
                }
            }
            else
            {
                if (c >= 'A' && c <= 'Z')
                {
                    c = c + 32;
                }
            }

            newMonth = newMonth + c;
        }
        return newMonth;
    }
    int FindDaysInMonth(string month, int year)
    {
        int days = 0;
        if (month == "January" || month == "March" || month == "May" ||
month == "July" || month == "August" || month == "October" ||
month == "December")
        {
            days = 31;
        }
        else if (month == "April" || month == "June" || month == "September"
||
month == "November")
        {
            days = 30;
        }
        else if (month == "February")
        {
            if ((year % 4 == 0 && year % 100 != 0) || year % 400 == 0)
            {
                days = 29;
            }
        }
        else
        {
            days = 28;
        }
        return days;
    }
}
```

```
void weatherReport(int days, int &hot, int &rainy, int &cloudy)
{
    char input;

    for (int i = 1; i <= days; i++)
    {
        cout << "Day " << i << " : Enter \"H\" for Hot, \"R\" for Rainy,
        \"C\" for Cloudy, (\"E\" to end): ";
        cin >> input;

        if (input >= 'A' && input <= 'Z')
        {
            input = input + 32;
        }

        if (input == 'e')
        {
            break;
        }
        else if (input == 'h')
        {
            hot = hot + 1;
        }
        else if (input == 'r')
        {
            rainy = rainy + 1;
        }
        else if (input == 'c')
        {
            cloudy = cloudy + 1;
        }
    }
else
{
    cout << "Invalid input! Please enter again." << endl;
i = i - 1;
}
}

void displayReport(int hot, int rainy, int
cloudy)
{
    cout << endl;
    cout << "Number of hot days this month: " << hot << endl;
    cout << "Number of rainy days this month: " << rainy << endl;
    cout << "Number of cloud days this month: " << cloudy << endl;

    cout << endl;
    cout << "The analysis star design as below:" << endl;
    cout << "-----" << endl;

    cout << "Hot    | ";
    for (int i = 1; i <= hot; i++)
    {
        cout << "* ";
    }
}
```

```

        cout << endl;

        cout << "Rainy | ";
        for (int i = 1; i <= rainy; i++)
        {

            cout << "* ";
        }
        cout << endl;

        cout << "Cloudy | ";
        for (int i = 1; i <= cloudy; i++)
        {

            cout << "* ";
        }
        cout << endl;

        cout << "-----" << endl; }

```

Output:

```

PS C:\Users\ianga\Downloads\Data Structure\C++ Codes> & "C:\Users\ianga\Downloads\Data Structure\C++ Codes\Lab2_PartC_Q2.exe"
Enter Your Month (e.g. August 2019): December 2005
Day 1 : Enter "H" for Hot, "R" for Rainy, "C" for Cloudy, ("E" to end): H
Day 2 : Enter "H" for Hot, "R" for Rainy, "C" for Cloudy, ("E" to end): H
Day 3 : Enter "H" for Hot, "R" for Rainy, "C" for Cloudy, ("E" to end): R
Day 4 : Enter "H" for Hot, "R" for Rainy, "C" for Cloudy, ("E" to end): R
Day 5 : Enter "H" for Hot, "R" for Rainy, "C" for Cloudy, ("E" to end): C
Day 6 : Enter "H" for Hot, "R" for Rainy, "C" for Cloudy, ("E" to end): C
Day 7 : Enter "H" for Hot, "R" for Rainy, "C" for Cloudy, ("E" to end): C
Day 8 : Enter "H" for Hot, "R" for Rainy, "C" for Cloudy, ("E" to end): C
Day 9 : Enter "H" for Hot, "R" for Rainy, "C" for Cloudy, ("E" to end): H
Day 10 : Enter "H" for Hot, "R" for Rainy, "C" for Cloudy, ("E" to end): H
Day 11 : Enter "H" for Hot, "R" for Rainy, "C" for Cloudy, ("E" to end): H
Day 12 : Enter "H" for Hot, "R" for Rainy, "C" for Cloudy, ("E" to end): H
Day 13 : Enter "H" for Hot, "R" for Rainy, "C" for Cloudy, ("E" to end): R
Day 14 : Enter "H" for Hot, "R" for Rainy, "C" for Cloudy, ("E" to end): R
Day 15 : Enter "H" for Hot, "R" for Rainy, "C" for Cloudy, ("E" to end): R
Day 16 : Enter "H" for Hot, "R" for Rainy, "C" for Cloudy, ("E" to end): R
Day 17 : Enter "H" for Hot, "R" for Rainy, "C" for Cloudy, ("E" to end): C
Day 18 : Enter "H" for Hot, "R" for Rainy, "C" for Cloudy, ("E" to end): C
Day 19 : Enter "H" for Hot, "R" for Rainy, "C" for Cloudy, ("E" to end): C
Day 20 : Enter "H" for Hot, "R" for Rainy, "C" for Cloudy, ("E" to end): C
Day 21 : Enter "H" for Hot, "R" for Rainy, "C" for Cloudy, ("E" to end): C
Day 22 : Enter "H" for Hot, "R" for Rainy, "C" for Cloudy, ("E" to end): C
Day 23 : Enter "H" for Hot, "R" for Rainy, "C" for Cloudy, ("E" to end): C
Day 24 : Enter "H" for Hot, "R" for Rainy, "C" for Cloudy, ("E" to end): H
Day 25 : Enter "H" for Hot, "R" for Rainy, "C" for Cloudy, ("E" to end): H
Day 26 : Enter "H" for Hot, "R" for Rainy, "C" for Cloudy, ("E" to end): R
Day 27 : Enter "H" for Hot, "R" for Rainy, "C" for Cloudy, ("E" to end): E

Number of hot days this month: 8
Number of rainy days this month: 7
Number of cloud days this month: 11

The analysis star design as below:
-----
Hot      | * * * * *
Rainy    | * * * * *
Cloudy   | * * * * *
-----

```

lab 33(13 August 2026)

Update for pA q1

Write a C++ program to calculate the average of three integers. Make four different functions of average() (refer to Figure 2) and display average in main() function. These function will / will not return any result back to the main() function.

Code:

```
#include <iostream>
using namespace std;

void averageNoReturnNoParameter();
void averageNoReturnWithParameter(int num1, int num2, int num3);
double averageWithReturnNoParameter();
double averageWithReturnWithParameter(int num1, int num2, int num3);

int main()
{
    int num1, num2, num3;
    double average;

    cout << "Enter three integers: ";
    cin >> num1 >> num2 >> num3;

    cout << endl;

    cout << "Function without return without parameters" << endl;
    averageNoReturnNoParameter();

    cout << endl;
    cout << "Function without return with parameters" << endl;
    averageNoReturnWithParameter(num1, num2, num3);

    cout << endl;
    cout << "Function with return without parameters" << endl;
    average = averageWithReturnNoParameter();
    cout << "Average = " << average << endl;

    cout << endl;
    cout << "Function with return with parameters" << endl;
    average = averageWithReturnWithParameter(num1, num2, num3);
    cout << "Average = " << average << endl;

    return 0;
}

void averageNoReturnNoParameter()
{
    int num1, num2, num3;
    double average;

    cout << "Enter three integers: ";
    cin >> num1 >> num2 >> num3;

    average = (num1 + num2 + num3) / 3.0;
```

```
        cout << "Average = " << average << endl;
    }

void averageNoReturnWithParameter(int num1, int num2, int num3)
{
    double average = (num1 + num2 + num3) / 3.0;

    cout << "Average = " << average << endl;
}

double averageWithReturnNoParameter()
{
    int num1, num2, num3;
    double average;

    cout << "Enter three integers: ";
    cin >> num1 >> num2 >> num3;

    average = (num1 + num2 + num3) / 3.0;

    return average;
}

double averageWithReturnWithParameter(int num1, int num2, int num3)
{
    double average = (num1 + num2 + num3) / 3.0;

    return average;
}
```

Output:

```
PS C:\Users\ianga\Downloads\Data Structure\C++ Codes> & "C:\Users\ianga\Downloads\Data Structure\C++ Codes\Lab33_Average.exe"
Enter three integers: 2 3 4

Function without return without parameters
Enter three integers: 2 3 4
Average = 3

Function without return with parameters
Average = 3

Function with return without parameters
Enter three integers: 2 3 4
Average = 3

Function with return with parameters
Average = 3
```